

# impl Sample for ZExpFamily<1>

Michael Shoemate (@Shoeboxam)

This proof resides in “**contrib**” because it has not completed the vetting process.

Proves soundness of the implementation of **Sample** for **ZExpFamily<1>** in **mod.rs** at commit **f5bb719** (outdated<sup>1</sup>).

## 1 Hoare Triple

### Precondition

#### Compiler-verified

None

#### Caller-verified

**self** represents a valid probability distribution.

### Pseudocode

```
1 class ZExpFamily1: # analogous to impl Sample for ZExpFamily<1> in Rust
2     def sample(self, shift):
3         return shift + sample_discrete_laplace(self.scale) #
```

### Postcondition

**Theorem 1.1.** Either returns **Err(e)** independently of the input **shift**, or **Ok(shift + Z)** where **Z** is a sample from the distribution defined by **self**.

*Proof.* By the precondition, since **self** represents a valid probability distribution, then by the definition of **ZExpFamily**, **self.scale** is non-negative.

Since the preconditions for **are** met (non-negative scale), then we claim the postcondition that the returned value follows a given distribution. This distribution matches that defined by **self**.

By the addition on line **3**, the implementation satisfies the postcondition. □

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<sup>1</sup>See new changes with `git diff f5bb719..9205d4f rust/src/measurements/noise/mod.rs`